

CUSTOMER RETAIL ANALYSIS USING MACHINE LEARNING

PROJECT OVERVIEW

This project focuses on analyzing a customer retail dataset by applying different Machine Learning algorithms and comparing their performance. The project helps in understanding data preprocessing, visualization, model building, prediction, and evaluation techniques.

AIM OF THE PROJECT

The main aim of this project is to develop and compare multiple Machine Learning models using customer retail data. Through this project, students can learn data cleaning, visualization, training models, testing predictions, and evaluating algorithm performance.

FEATURES TAKEN FROM DATASET

- Quantity Purchased
- Product Price
- Customer Country

MACHINE LEARNING ALGORITHMS APPLIED

- Logistic Regression Model
- Decision Tree Algorithm
- K-Nearest Neighbors (KNN) Algorithm

STEP-BY-STEP WORKFLOW

1. Import customer retail dataset using Pandas.
2. Perform preprocessing and handle missing data.
3. Convert categorical data into numerical format using Label Encoding.
4. Create visualizations using Matplotlib.
5. Split the dataset into training and testing data.
6. Train Logistic Regression classifier.
7. Train Decision Tree classifier.
8. Train KNN classifier.
9. Evaluate models using Accuracy Score and Confusion Matrix.

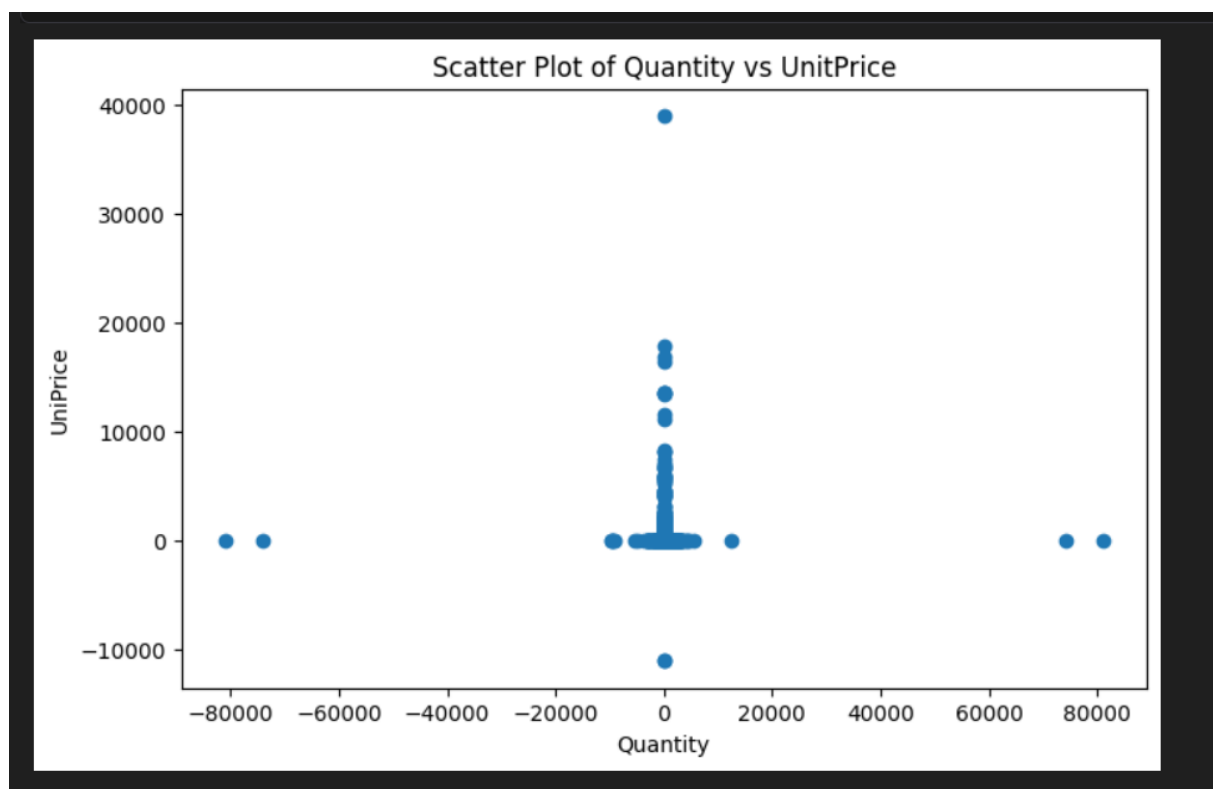
10. Compare the performance of all algorithms with graphs and charts.

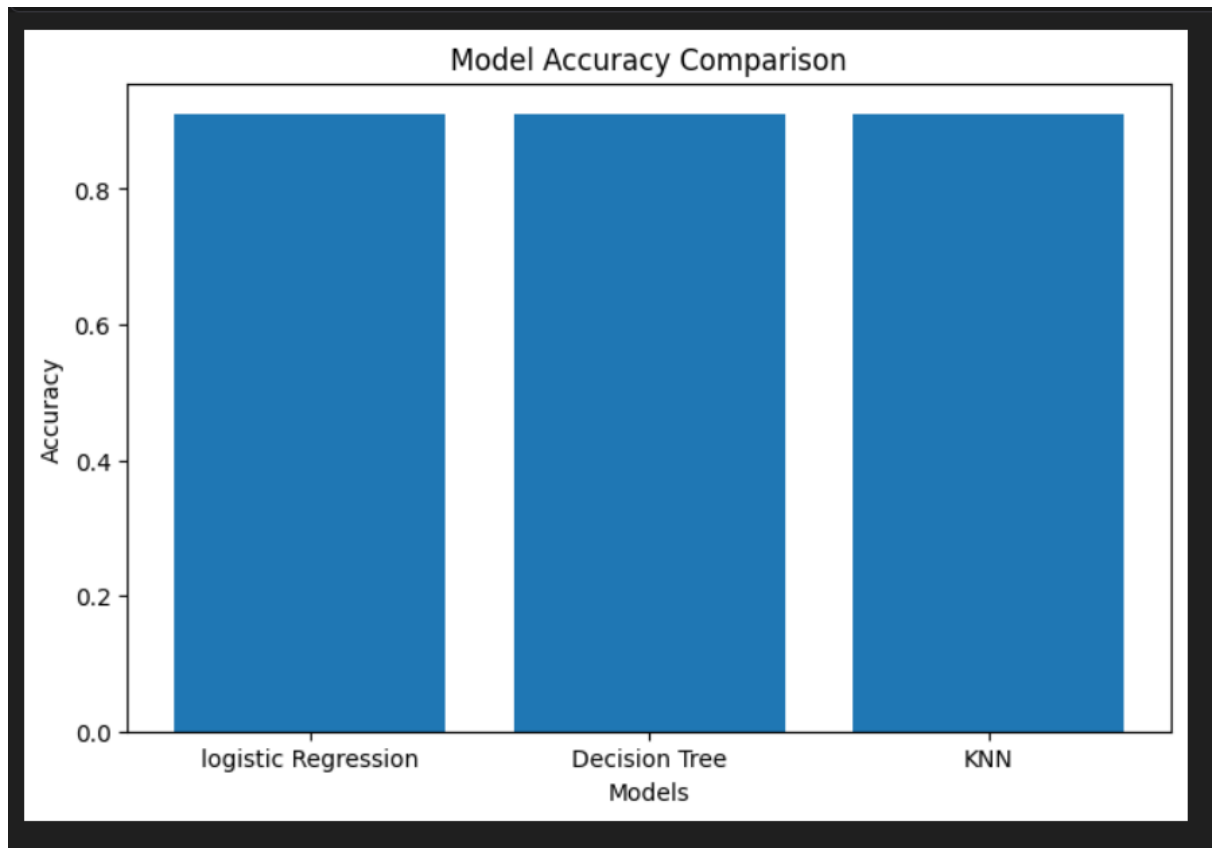
PERFORMANCE EVALUATION METHODS

- Accuracy Score
- Confusion Matrix

TOOLS AND LIBRARIES USED

- Python
- Pandas
- NumPy
- Matplotlib





CONCLUSION

This project demonstrates how different Machine Learning models can be used on customer retail datasets for prediction and comparison. After evaluation, the performances of various algorithms are compared to identify the most accurate model.